



Research Profile

My research tackles a fundamental bottleneck in the design-through-analysis pipeline: turning a CAD model into a discretization that is geometrically faithful, mathematically well-posed, and fast to solve. Working at the interface of **computer-aided design (CAD)** and **computer-aided engineering (CAE)**, I develop **isogeometric analysis (IGA)** methods—provably injective and regular parameterizations, PDE-based construction, and accelerated solvers—that keep geometry, discretization, and simulation consistent throughout. This work has moved from theory into practice: it now underpins industrial-grade structured meshing for twin-screw compressors, semi-analytical thermal simulation for metal additive manufacturing, and, most recently, isogeometric formulations of physics solvers such as the lattice Boltzmann method.

Research Highlights

- **Analysis-suitable parameterization for IGA:** optimization-based and PDE-based volumetric constructions with quality control for complex domains.
- **Geometry-to-simulation pipelines:** boundary parameter matching (incl. Schwarz–Christoffel tools) and multi-sided domain parameterizations enabling reliable IGA workflows.
- **Solver acceleration:** preconditioned Anderson acceleration to improve efficiency and robustness of parameterization PDE solvers.
- **Industrial-grade spline meshing:** spline-based structured mesh generation for screw-machine simulations (SplineMesh / SCORG™).
- **Extending IGA beyond structural/thermal analysis:** ongoing work on physics-based extensions, including an isogeometric lattice Boltzmann method (IGA-LBM) for body-fitted, CAD-exact fluid simulation on curved geometries (presented at WCCM-ECCOMAS 2026).

Academic Positions

01/2024–
Present **Postdoctoral Researcher**, *Delft University of Technology*, Department of Applied Mathematics (Numerical Analysis), The Netherlands.

Education

2019–2023 **Ph.D. Mathematics**, *Dalian University of Technology*, School of Mathematical Sciences, China.

2017–2019 **M.Sc. Mathematics**, *Dalian University of Technology*, School of Mathematical Sciences, China.

2013–2017 **B.Sc. Mathematics**, *Dalian University of Technology*, School of Mathematical Sciences, China.

Research Experience

10/2021–
12/2023 **Visiting Ph.D. Researcher**, *Delft University of Technology*, Department of Applied Mathematics (Numerical Analysis), The Netherlands.

Research Projects

01/2024–
12/2026 **Advanced Geometric Modelling and Simulation Techniques for Complex Engineering Systems**. Postdoctoral Fellowship. *Role:* PI.

10/2021–
10/2023 **PDE-Based Parameterization Method for Isogeometric Analysis and Its Application in Twin-Screw Rotary Compressors**. China Scholarship Council (CSC). *Role:* PI.

01/2021–
12/2024 **Theoretical Study and Application of Parametric Surfaces/Volumes in Isogeometric Analysis**. NSFC, General Program. *Role:* Participant.

01/2017–
12/2020 **Geometric Properties of Parametric Curves and Surfaces**. NSFC, General Program. *Role:* Participant.

Selected Peer-reviewed Publications

- 26 refereed journal papers, 2 refereed conference papers, 1 book chapter; full list at jiyess.github.io/publications.
- 2027 J. Li, **Y. Ji**, H. Verhelst, H. den Besten, M. Möller. **Parameterization-driven arbitrary Lagrangian–Eulerian method for large-deformation isogeometric fluid–structure interaction**. *Computer Methods in Applied Mechanics and Engineering*, 463, 119358 (2027).
- 2026 J. Cao, **Y. Ji**, M. Möller, C.-G. Zhu. **Extended r-adaptive isogeometric analysis for weak-discontinuous problems**. *Computer-Aided Design*, 198, 104108 (2026).
- 2026 Y. Yang, **Y. Ji**, M. Möller, C. Ayas. **Efficient thermal simulation in metal additive manufacturing via semi-analytical isogeometric analysis**. *Computer Methods in Applied Mechanics and Engineering*, 457, 118992 (2026).
- 2025 Y. Yang, **Y. Ji**, M. Möller, C. Ayas. **Computational efficient process simulation of geometrically complex parts in metal additive manufacturing**. *International Journal of Heat and Mass Transfer*, 248, 127059 (2025).
- 2025 L. Yang, W. Wang, **Y. Ji**, C.-G. Zhu, C. C. L. Wang. **Space–time isogeometric topology optimization with additive manufacturing constraints**. *Computer Methods in Applied Mechanics and Engineering*, 441, 117976 (2025).
- 2024 **Y. Ji**, M. Möller, Y.-Y. Yu, C.-G. Zhu. **Boundary parameter matching for isogeometric analysis using Schwarz–Christoffel mapping**. *Engineering with Computers*, 40(6), 3929–3947 (2024).
- 2024 **Y. Ji**, M. Möller. **Yet another structured mesh generator for screw machines simulation**. *IOP Conference Series: Materials Science and Engineering (ICSM 2024)*, 1322(1), 012014 (2024).
- 2023 **Y. Ji**, K.-W. Chen, M. Möller, C. Vuik. **On an improved PDE-based elliptic parameterization method for isogeometric analysis using preconditioned Anderson acceleration**. *Computer Aided Geometric Design*, 102, 102191 (2023). (GMP 2023 Best Paper Award)
- 2023 **Y. Ji**, M.-Y. Wang, Y.-Y. Yu, C.-G. Zhu. **Curvature-based r-adaptive isogeometric analysis with injectivity-preserving multi-sided domain parameterization**. *Journal of Systems Science and Complexity*, 36(1), 53–76 (2023).
- 2022 **Y. Ji**, M.-Y. Wang, Y. Wang, C.-G. Zhu. **Curvature-based r-adaptive planar NURBS parameterization method for isogeometric analysis using bi-level approach**. *Computer-Aided Design*, 150, 103305 (2022).
- 2022 **Y. Ji**, M.-Y. Wang, M.-D. Pan, Y. Zhang, C.-G. Zhu. **Penalty function-based volumetric parameterization method for isogeometric analysis**. *Computer Aided Geometric Design*, 94, 102081 (2022).
- 2021 **Y. Ji**, Y.-Y. Yu, M.-Y. Wang, C.-G. Zhu. **Constructing high-quality planar NURBS parameterization for isogeometric analysis by adjustment control points and weights**. *Journal of Computational and Applied Mathematics*, 396, 113615 (2021).
- 2021 Y.-Y. Yu, **Y. Ji**, J.-G. Li, C.-G. Zhu. **Conditions for injectivity of toric volumes with arbitrary positive weights**. *Computers & Graphics*, 97, 88–98 (2021). (CAD/Graphics 2021 Best Paper Award)

Selected Talks

Delivered 30+ invited and conference presentations; selected highlights below.

- Sep. 2026 **Oral**, *SplineMesh v3.0: Robust spline-based structured mesh generation for positive-displacement rotary machines with sharp features*, ICSM 2026, Dortmund, Germany.
- Jul. 2026 **Invited Talk**, *CAD-CAE integration: From theory and algorithms to engineering practice*, CSIAM GDC 2026, Hohhot, China.
- Jul. 2026 **Oral**, *An isogeometric formulation of the lattice Boltzmann method on CAD-exact geometries*, WCCM-ECCOMAS 2026, Munich, Germany.
- Apr. 2026 **Invited Talk**, *Analysis-suitable parameterization for isogeometric analysis*, AROMATH Team, Inria Centre at Université Côte d’Azur, Sophia Antipolis, France.

- Oct. 2025 **Invited Talk**, *Analysis-suitable parameterization for isogeometric analysis*, Institut für Leichtbau und Struktur-Biomechanik, TU Wien, Vienna, Austria.
- Sep. 2025 **Oral**, *IGA-LBM: Isogeometric lattice Boltzmann method*, IGA 2025, Eindhoven, Netherlands.
- Sep. 2025 **Short Course**, *SplineMesh v2.0: Spline Mesh technology for twin-screw compressors*, 7th Short Course & Forum on CFD in Rotary Positive Displacement Machines, London, UK.
- Sep. 2024 **Oral**, *Yet another structured mesh generator for screw machines simulations*, ICSM 2024, Dortmund, Germany.

Teaching, Mentoring, and Supervision

- Lecturer Co-developer and lecturer of a graduate special course on **Isogeometric Analysis** at TU Delft (Q3–Q4 2026), jointly prepared with Prof. Matthias Möller (TU Delft) and Prof. Stefanie Elgeti (TU Wien).
- Supervision Supervision/co-supervision of Ph.D. and Master’s research in IGA and spline geometry (e.g., semi-analytical IGA for thermal/process simulation in metal additive manufacturing; multi-sided parameterization; extended IGA; spline approximation).

Research Software and Infrastructure

- **G+Smo (Geometry + Simulation Modules, <https://github.com/gismo/gismo>)**: member of the core development team of the open-source C++ IGA library, supporting geometry processing, discretization, and simulation workflows.
- **SplineMesh / SCORG™**: lead developer of spline-based structured mesh generation modules for industrial modeling (screw machines), focusing on robustness, quality, and integration with simulation pipelines.

Service and Professional Activities

- Refereeing Reviewer for *Computer-Aided Design*, *Computer Aided Geometric Design*, *Scientific Reports*, *Engineering with Computers*, *Advances in Engineering Software*, *Finite Elements in Analysis & Design*.
- 2022–Now Reviewer for **Mathematical Reviews**, American Mathematical Society.
- 2020–Now Life member, China Society for Industrial and Applied Mathematics (CSIAM).

Honors and Awards (Selected)

- 07/2026 Conference Best Paper Award **and** Best Poster Award, CSIAM GDC 2026 (CSIAM Conference on Geometric Design and Computing), Hohhot, China.
- 08/2025 Conference Best Paper Award, CSIAM GDC 2025, Yantai, China.
- 10/2024 Conference Best Paper Award, ICSM 2024 (International Conference on Screw Machines), Dortmund.
- 12/2023 Outstanding Graduate of Liaoning Province (Class of 2024).
- 07/2023 Conference Best Paper Award, GMP 2023, Genova, Italy.
- 02/2023 Challenge Winner, Amazon Web Services Challenge at the SIAM Hackathon 2023, Amsterdam, Netherlands.
- 2022 & 2016 National Scholarship (twice; Ph.D. 2022 and undergraduate 2016), Ministry of Education of China (awarded to approximately 0.2% of students nationwide).
- 05/2021 Conference Best Paper Award, CAD/Graphics 2021, Xi’an, China.

Skills

- Programming C/C++ (modern CMake-based projects), Python (scientific computing), MATLAB, Julia, L^AT_EX.
- Numerical Methods Partial Differential Equations, Finite Element Methods, Isogeometric Analysis, Structured Grid Generation.
- Geometry & Modeling B-splines and NURBS, Parametric Curves/Surfaces, Mesh Generation, Geometric Mapping and Regularity Analysis.
- Languages Mandarin (native), English (fluent).